

xMCF

ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues [#105](#) / [#61](#)
2026-01-28

Agenda

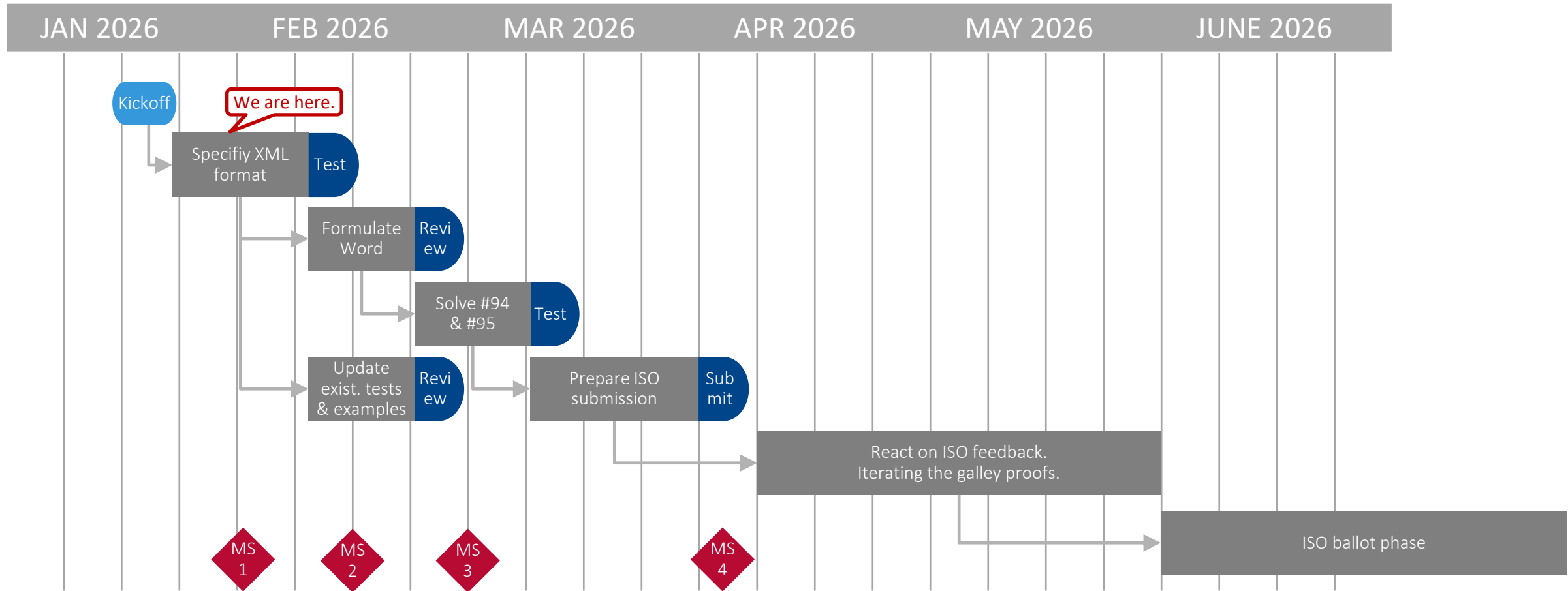
As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-01-28_VideoConference/README.md

Today's objective: **Create XML format suggestions**

- Welcome
- Timeline for next release
- Focus on detailing the XML elements, based e.g. on the [comment in GitHub](#) at issue [#105](#).
- Steps we should take next



Timeline for a release in the summer



Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.
Last week's meeting of the Policy and Planning Committee (PPC) was postponed to 18th of Feb.

Detailing XML Elements — Template Syntax, Screw Example

We identified the example XML file [chapter 9 5 7 exampleA patched for xMCF3.2.xml](#) (provided by the comment of [2026-10-14 09:17+0100](#)) as a good starting point.

It suggests as a syntax for defining an (file-embedded) list of *connection templates* at XML root level:

```
<connection_templates>
  <connection_0d label="screw_template_4711">
    <threaded_connection length="50." diameter="10"
      head_diameter="16.0" head_height="5" sink_size="4" thread_length="35"
      torque="80" angle="30" pretension="1200" part_code="M10x50 12.9" >
      <screw /> <!-- Screw may come without any attributes -->
      <washer outer_diameter="20"/>
    </threaded_connection>
  </connection_0d>
</connection_templates>
```

Homework: Consider what “misuse” or contradictory set of data can occur by this approach!

Homework: Create more examples, e.g. for gum drops, spot welds, rivets, `<screw/>` with attributes. → today’s meeting directory in GitHub
2026-01-25: Created examples for spotweld, rivet, gum drop, and gum drop sequence in [2026-01-28 VideoConference](#).

Detailing XML Elements — Reference Syntax, Screw Example

The example suggests as syntax for *referring to* a connection template at `<connection_0d/>` level:

Default form — no change to / overwriting template data:

```
<!-- Screw exactly as defined in connection_templates: -->
<connection_0d label="SCREW_100532" template="screw_template_4711">
  <threaded_connection>
    <normal_direction x="2.9" y="0.1" z="0.0" />
  </threaded_connection>
  <loc> 1500.3809 838.75885 730.6529 </loc>
</connection_0d>
```

Minimal change — overwriting an attribute value:

```
<!-- Screw exactly as defined in connection_templates, but with smaller torque: -->
<connection_0d label="SCREW_100533" template="screw_template_4711">
  <threaded_connection torque="70" >
```

Homework: Move the reference to the template to the `<connection_0d/>`-level! (Done, 2026-01-25)

```
    <normal_direction x="3.0" y="0.0" z="0.0" />
  </threaded_connection>
  <loc> 1540.3809 838.75885 730.6529 </loc>
</connection_0d>
```



Detailing XML Elements — Reference Syntax, Screw Example

The example suggests as syntax for *referring to* a connection template at `<connection_0d/>` level:

Overwriting a child element:

```
<!-- Screw exactly as defined in connection_templates, but with larger washer: -->
<connection_0d label="SCREW_101537" template="screw_template_4711">
  <threaded_connection>
```

Homework: Move the reference to the template to the `<connection_0d/>`-level! (Done, 2026-01-25)

```
    <normal_direction x="3.1" y="-0.1" z="0.0" />
    <washer outer_diameter="30"/>
  </threaded_connection>
  <loc> 1580.3809 838.75885 730.6529 </loc>
</connection_0d>
```

Attention: Here, only the `outer_diameter` of the `<washer/>` is overwritten. Any other washer-parameter given from the template stays effective.

Detailing XML Elements — Template Syntax, Bolt Example

The example extends to cases of *deeper structured* elements:

```
<connection_templates>
  <connection_0d label="bolt_template_4712">
    <threaded_connection length="50" diameter="10"
      head_diameter="16" head_height="5" thread_length="35"
      torque="80" angle="30" pretension="1200" part_code="M10x50 12.9">
      <normal_direction x="0" y="0" z="-10"/>
      <washer outer_diameter="20" inner_diameter="10.3" thickness="1.5" attached="true"/>
      <bolt>
        <nut diameter="16." height="5" static_friction="0.8" clipped_to="4"/>
      </bolt>
    </threaded_connection>
  </connection_0d>
</connection_templates>
```

Rename “internal_catalog” to “~~embedded_catalog~~” “connection_templates” and attribute name “internal_catalog_item” to “template”! (Done, 2026-01-25)

Detailing XML Elements — Reference Syntax, Bolt Example

The connection template can be *referred to* for example like this:

```
<!-- Bolt exactly as defined in connection_templates, but with less torque, larger washer & nut: -->
<connection_0d label="BOLT_100732" template="bolt_template_4712">
  <threaded_connection torque="70">
    <normal_direction x="3.1" y="-0.1" z="0.0"/>
    <washer outer_diameter="30"/>
```

Check, whether a washer may refer to a connection template, too! → we need examples to better understand.

```
    <bolt>
      <nut height="8"/>
```

Check, whether a nut may refer to a connection template, too! → we need examples to better understand

```
    </bolt>
  </threaded_connection>
</connection_0d>
```

Rule (→ Word file): If the connection template is a `<screw/>`, the referring element must not redefine it to a `<bolt/>` and vice versa!

To be extended to similar situations in case of future extensions of χMCF, e.g. a spotweld (from template) must not be redefined to a gumdrop!

Detailing XML Elements — Template Syntax, Gumdrops Example

(new — see file “chapter_10_5_exampleB__with_templates.xml”)

For a gumdrop, the example template would be:

```
<connection_templates>
  <connection_0d label="GumDrop template">
    <!-- Assumed Unit system with mass attribute with value="kg" -->
    <gumdrop diameter="5.0" mass="0.0033" material="Glue_Material" />
  </connection_0d>
</connection_templates>
```



Detailing XML Elements — Reference Syntax, Gumdrops Sequence (new — see file “chapter_10_5_exampleB__with_templates.xml”)

For a gumdrop sequence, the example reference could be (as suggested in [GitHub comment](#)):

```
<!-- Sequence of gum drops exactly as defined in connection_templates: -->
<connection_1d label="DROP_LINE_11000" template="GumDrop template">
  <sequence_connection_0d spacing="30.0" margin="1.0" priority="spacing">
    <gumdrop/>
  </sequence_connection_0d>
  <loc_list>
    ...
  </loc_list>
</connection_1d>
```

Observation (CF): A template for a 0d item as parameter of a 1d item appears inappropriate to me.

What about this approach:

```
<connection_1d label="DROP_LINE_11000" template="Later, here could be place for a sequence template.">
  <sequence_connection_0d spacing="30.0" margin="1.0" priority="spacing" template="GumDrop template">
```

Contentual Action Items $\triangleq$ Next Steps:

▪ Standard document (Word file):

- Description of overall concept & objectives, intended application & motivation etc.
- Description of semantics for internal catalog
- Description of semantics for reference to a connection template
- Definition of example use cases and in-document XML-examples

▪ XML & XSD — See discussion in [#105](#)!

- Sketching XML example file(s), incl. (w.i.p.)
 - precedence rule (w.i.p.) and dropped mandatory attribute,
 - internal (w.i.p.) and external file-based catalog
- Description of syntax for internal catalogs (w.i.p.)
- Description of syntax for reference to a connection template:
 - internal (w.i.p.), external file-based, external database-based

- Drafting XML schema (XSD) and test cases / test example files

Homework: Collect use cases which can yield e.g. test cases, shared as Markdown files in GitHub at https://github.com/economidis-nick/createXSDforxMCF/tree/issues_61_94_95_105/VDA_FAT_AK_25/Meetings/2026-01-28_VideoConference

- Investigate xMCF elements and attributes for eligibility for template.
⇒ we need a rule, here! ⇒ Homework!



Next meetings

Conclusion from this discussion: ...

- 2026-02-04, 15:00 CET — full group (new time slot) to be repurposed to focus group
— objective: agree on XML format suggestions
- 2026-02-11, 15:00 CET — focus group on [#105](#) / [#61](#) (repurposed time slot)
— objective: create DOC suggestions & examples
- 2026-02-18, 15:00 CET — full group (new time slot)
— objective: agree on DOC suggestions & examples to be changed to XML format

CF sent / will send out invitations.

